

Maurice Ngicho

Backend Software Engineer | Python (Flask) • Node.js • SQL • Rest API

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GitHub: <https://github.com/MauriceNgicho> | Portfolio: <https://mauricengicho.github.io/Portfolio>

Summary

Backend software engineer with strong experience in designing and implementing scalable, secure backend systems. Skilled in Python (Flask, FastAPI), Node.js, and SQL, with hands-on expertise in RESTful and GraphQL API development, database optimisation, authentication/authorisation, and testing. Graduate of ALX Software Engineering program with proven ability to collaborate in cross-functional teams and deliver production-ready solutions that enhance performance, maintainability, and user experience.

Skills

Languages: Python(Flask, FastAPI), [Node.js](#), JavaScript, TypeScript, SQL, Shell

APIs & Databases: RESTful APIs, PostgreSQL, MySQL, MongoDB, Redis, ORMs(SQLAlchemy)

Testing & Tools: Unit/Integration Testing, Mocha/Chai, Git

Tools & DevOps: Git/GitHub, Postman, Flask-JWT-Extended

Cloud & Deployment: Railway, Render, Vercel,

Security: Authentication & Authorisation, JWT, role-based access control

AI-Enhanced Development: GitHub Copilot, ChatGPT, Prompt Engineering

Projects

WaterTek – Smart Utility Billing System(Flask + MySQL)

- Implemented JWT authentication with role-based access(user/admin).
- Designed and optimised SQL queries for billing accuracy and efficiency
- Built modules for billing, meter reading, and issue tracking.
- Added pagination, relational linking of users to bills, and performance optimisation for scalability

SkillLink — Corporate Learning Management System(Flask + MySQL)

- Developed user authentication with refresh tokens and blacklisting.
- Designed REST APIs for courses, user progress, and admin controls.

- Integrated Redis for session/token management.

Files Manager — File storage and management API([Node.js](#) + MongoDB + Redis)

- Built a token-based user authentication system
- File CRUD API with Base64 login, Redis session tokens, and access control.
- Used Redis for user session management and rate limiting

Education

ALX Software Engineering Program

Remote | Graduated

Project-based full-time program focused on backend development, system design, databases, and DevOps.

Diploma in Electrical and Electronics Engineering (Telecommunication Option)

Kenya Institute of Mass Communication | Graduated

St. Gabriel's Lanet High School

Graduated

Certifications

- ALX Software Engineering Certificate
- Python for Everybody – Simplilearn
- ALX AiCE – AI Career Essentials

Experience

Technician — Sai Offices | May 2025 – Present

Maintained IT and office infrastructure, ensuring uptime of systems and supporting team operations.

Technician — MFI Document Solutions | Sep 2022 – Apr 2025

Supported clients with office systems, providing troubleshooting and user training.